Memory Is Not Enough: Separating Drift from Background Conflict in Continual Document Understanding

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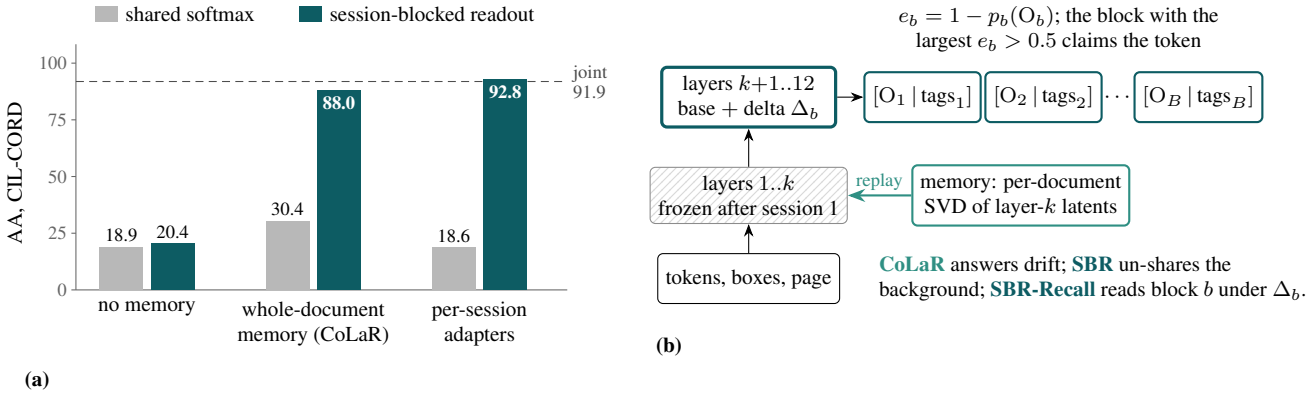


Figure 1. **Two causes of forgetting, and a remedy matched to each.** (a) Class-incremental CORD, LayoutLMv3, AA under session-restricted scoring; bars are the cell means of Tab. 2, which gives the seeds and protocol of each cell. A whole-document memory alone and a session-blocked readout alone both stay far below joint training; together they come within a few points of it, and per-session low-rank adapters with a blocked readout (SBR-Recall) are on par with it without a memory, while a different adapter method with a shared softmax (CL-LoRA) collapses. (b) Layers 1..k are frozen after the first session; the memory stores per-document SVDs of layer-k activations and re-grounds the drifting upper layers (CoLaR); each session gets its own background logit and tokens are routed with no task identity (SBR); SBR-Recall evaluates each block under the trunk delta of the session that created it.

Abstract

Continual learning for document understanding is commonly treated as a single problem, catastrophic forgetting, addressed by a single remedy, replay. We show that continual key-information extraction instead fails for two separable reasons. The first is drift *of the output-facing computation, which arises in every scenario and on all four document encoders we study; freezing the affected layers relocates the drift rather than removing it, and none of the five method families we evaluate that update a shared network without old-task supervision recovers from it. The second,* background conflict*, arises only when the label space grows: current and stored documents supervise the shared background class inconsistently, an effect that no memory can repair and that the standard per-session evaluation conceals. A factorial analysis on class-incremental CORD separates the two causes: a whole-document memory and a session-blocked readout are each insufficient in isolation, whereas together they attain* 88.0 *average F1 against* 91.9 *for joint training. We propose a remedy for each cause.* ***CoLaR*** *(Compressed Latent Replay) replays a truncated SVD of each stored document's mid-network activations and matches raw-exemplar replay under domain shift without storing raw pixels or token identifiers.* ***SBR*** *(Session-Blocked Readout) assigns each session its own background logit and routes every token without task identity. Finally,* ***SBR-Recall*** *replaces the memory with per-session low-rank trunk updates when label growth itself supplies the routing key, and reaches* 92.8*, on par with joint training.*

1. Introduction

Organisations process a continuous stream of visually-rich documents (forms, invoices, receipts, contracts) whose meaning depends on the words and on where they sit on the page. Key-information extraction (KIE) is cast as token-level sequence labelling over text, 2-D layout and the page image, and multimodal encoders such as LayoutLMv3 [12], LiLT [37] and BROS [10] define the state of the art. Deployed systems are not static: new document types appear, schemas grow and the input drifts, while retraining on all history is blocked by cost and by privacy constraints on earlier data. This is continual learning (CL), whose central obstacle is catastrophic forgetting [24].

The CL toolbox (regularisation [14], distillation [18], replay [3, 32], prompts [34, 41], low-rank adapters [39]) treats forgetting as one phenomenon with one dial between stability and plasticity. On documents that picture produces a puzzle. Under a domain shift with a fixed label space, plain replay comes close to joint training on every backbone we test. When the label space grows, every method, replay included, stays far below joint training, and a better memory helps little. The usual reading is that class-incremental learning is simply a harder case of the same forgetting.

We show it is a different failure. Continual KIE fails for two *separable* reasons. **Drift**: the output-facing computation (upper layers and classifier) moves when it is trained on the next task, and on a fixed label space every method we tested that keeps updating a shared network without old-task supervision loses the old readout. **Background conflict**: under label growth a token labelled "O" means "none of the classes seen so far", so current documents label old entities as background while stored documents label new entities as background, and both push the same logit in opposite directions on the same surface forms. Memory addresses the first and is blind to the second. The standard per-session scoring hides the second as well, because it charges a correct prediction of another session's entity as an error: a jointly trained model scores 33 under it and 91.9 once each error is charged once.

Figure 1a is the paper in one picture. On class-incremental CORD a whole-document memory (CoLaR, our compressed latent replay) with a shared softmax reaches 30.4 and a session-blocked readout (SBR) without memory reaches 20.4, both far closer to naive fine-tuning (18.9) than to joint training; the two together reach 88.0 ± 2.0, against 91.9 ± 1.3 for joint training. Neither remedy is sufficient alone and each cause needs its own remedy, which is what two separable causes predict.

Contributions.

1. **A corrected protocol for token-level class-incremental evaluation.** Session-restricted scoring charges every error at most once; we add a mask-free full-label F1, reported for clean-protocol runs as they complete, and show that the reported "floor" of class-incremental document IE is largely a scoring artefact (Sec. 3).
2. **A separation of two causes of forgetting.** Drift is present in every scenario, shared by four encoders, relocated rather than removed by freezing, and not repaired by five families of shared-network methods; background conflict is specific to label growth and survives a memory with perfect labels. A 2×2 over memory and readout isolates them (Sec. 4).
3. **SBR and SBR-Recall.** The Session-Blocked Readout (SBR) gives each session its own background logit and routes every token with no task identity; with a whole-document memory it comes within a few points of joint training on class-incremental CORD and reaches it on a mixed stream. SBR-Recall stores a low-rank trunk delta per session and reads each block under its own delta, which removes drift by construction and needs no memory on a single-schema stream (Secs. 5.2 and 5.3).
4. **CoLaR** (Compressed Latent Replay)**.** A falsification ladder shows that a document memory must keep whole documents, and that each document can be compressed on its own: per-document truncated SVD of mid-network activations matches raw-exemplar replay under domain shift at 60 MB (Sec. 5.1).

2. Related Work

Continual-learning families. Following Wang et al. [38], remedies fall into regularisation (EWC [14], SI [44], MAS [1], LwF [18]), replay (GEM [21], ER [32], DER++ [3], iCaRL [31]), optimisation geometry (GPM [33], OGD [5], C-Flat [2]), methods over a frozen backbone (L2P [41], DualPrompt [40], CODA-Prompt [34], O-LoRA [39], CL-LoRA [9], RanPAC [25]) and architectural isolation [22, 43]. Replay is empirically the strongest [23, 38] but stores raw data. Exemplar-free alternatives replay class-conditional feature statistics (PASS, FeTrIL, FeCAM [6, 28, 52]) or merge models and realign the classifier (LCA [35]).

Class-incremental learning as prediction plus task identification. Kim et al. [13] decompose class-incremental learning into within-task prediction and task-identity prediction, and tie the latter to out-of-distribution detection; recent adapter methods add one module per task and need no task identity at test time, combining their features (EASE [51]) or folding them into the weights (InfLoRA [19]). SBR instantiates the decomposition by predicting task identity per token through the per-session background logit, with a property specific to token-level extraction: each session's documents contain true-negative tokens for its classes, so this logit is

a task-identity detector trained on real negatives. In continual NER, ICE [20] trains per-session classifiers on a frozen encoder and SpanKL [48] uses per-type span heads with distillation; neither replays stored examples.

Latent and compressed replay. REMIND [8] stores product-quantised mid-level CNN feature maps per image, Latent Replay [27] each example's activations at a chosen layer, and generative replay [36] samples hidden features from one class-conditional generator. The first two keep one entry per example, as CoLaR does, but each example has one label; a document has one label per token bound to a position, which must be kept, and CoLaR compresses it by truncated SVD, not product quantisation (Sec. 5.1).

Where forgetting lives; continual IE. Ramasesh et al. [30] localise forgetting in image classifiers to the layers nearest the output, and Zhai et al. [45] find multimodal LLMs forget in fine-tuning even with a frozen image encoder; our drift analysis agrees with the former and adds the intervention. Continual NER handles the shift of "O" with distillation and pseudo-labels (CPFD [46]), causal-effect distillation (CFNER [50]) or entity-triplet generation [42]; continual relation extraction uses episodic memory with prototypes [4, 7, 49]. All are sentence-level and text-only. Document CL is nascent: GDI-Bench [17] freezes most parameters of a multimodal LLM by update size to limit forgetting; continual table detection [26] uses plain ER; ProtoNER [16], closest to ours, adds KIE classes to a LayoutLM-family model from few documents with prototypes. None separates causes of forgetting or studies the "O" class.

3. Setup

Task and scenarios. A document is a triple of OCR tokens, normalised boxes and the page image; the model emits BIO tags per token and is scored by entity-level F1 (`seqeval`). We use FUNSD (149/50 train/test forms), CORD (800/100/100 receipts, 30 classes) and SROIE (626/347 receipts). **DIL**: FUNSD→SROIE→CORD under one fixed four-entity schema; only the input distribution shifts. **CIL-CORD**: CORD's 30 classes arrive six at a time over five sessions; the head grows, and tokens of classes not yet introduced are labelled "O". **Mixed**: six sessions alternating new classes within CORD with new domains (FUNSD, SROIE) and ending in a FUNSD revisit. Details are in the supplement (Sec. A).

Backbones and methods. LayoutLMv3-base [12] is the primary encoder; LiLT [37], BROS [10] and text-only BERT are secondary. Heads are linear. The comparison set is fixed in advance: naive fine-tuning and joint training as references, and one published method per family: ER [32] and DER++ [3], LwF [18], CODA-Prompt [34], CL-LoRA [9], LCA [35], and the continual-NER methods CPFD [46] and IS3 [29]. Replay baselines keep 200 raw documents (DER++ $\alpha{=}\beta{=}0.5$); in clean-protocol cells ER instead keeps as many raw documents as our memory. LwF uses $T{=}2$ and CL-LoRA rank-8 adapters on the attention projections. Our memory keeps 50 documents per session (per task on DIL).

Training protocol. AdamW, learning rate $5{\times}10^{-5}$, weight decay 0.01, gradient clipping 1.0, batch size 2, three seeds. Under the *clean protocol* every session trains in full precision for at most 10 epochs (joint training: 100) with early stopping, patience 2, on a 10% split held out from that session's *training* data; the test split is never used for model selection. Cells marked † come from earlier runs (the *legacy protocol*), which differ from the clean protocol in three ways. Every legacy cell early-stops on the session's *test* split, except joint training on CIL-CORD and Mixed, which holds out 10% of its pooled training set. Epoch caps differ by method: 100 for naive, joint training, ER, DER++, LwF, CPFD and IS3, 30 for CODA-Prompt and CL-LoRA, 10 for the CoLaR-family rows and LCA. Precision differs too: most re-scored CIL-CORD and Mixed cells ran in fp16 mixed precision, but IS3 and LCA always ran in fp32, and seed 42 of ER, DER++, LwF, CPFD, CL-LoRA and CODA-Prompt on CIL-CORD ran in fp32, so several cells mix precisions across seeds; the CIL-CORD cells of CoLaR-SBR, no-memory SBR, SBR-Recall and LexSlot, and the DIL SBR cells, are fp32. In Tab. 2 the shared-softmax cells are therefore fp16 or mixed and the blocked-readout cells fp32. The DIL rows of our first grid (Tab. 1) saved no configuration; their settings are the grid defaults and could not be confirmed per run. The supplement (Sec. D) lists the settings of every † cell. A cell never combines the two protocols.

Scoring. In CIL-CORD and the class-incremental halves of Mixed, session j's test labels show only session j's classes; every other entity is "O". Scoring raw predictions against them (the practice we call legacy scoring) charges a model that *correctly* labels another session's entity with a false positive in every other session, so a jointly trained model reads 33. *Session-restricted scoring* maps a predicted tag outside session j's label set to "O" before span matching, without re-taking the argmax among the allowed tags. Each predicted entity is then scored in the one session that owns its class, so an error is charged at most once and a confusion between sessions still costs a miss. Because a session's test split omits receipts with none of its entities, we also report a mask-free *full-label F1*: the final model on the unmasked CORD test set with all 30 classes. Only clean-protocol runs record it, so Tab. 3 gives it for the runs completed so far, with the number of seeds marked. From $R[i, j]$ (F1 on session j after training session i) we report av-

Table 1. **Drift and its remedy are shared by four encoders** (DIL, AA, mean ± s.d. over three seeds, all cells legacy protocol†, from our first grid, whose per-run settings were not saved). Methods without old-task supervision stay at the naive level; on average replay comes within one point of joint training (the reference) or exceeds it, on every backbone. "–": did not fit the 6 GB GPU.

Backbone	Naive	EWC	LwF	ER	DER++	Joint
LayoutLMv3	41.3±1.5	41.2±1.1	43.7±1.1	87.9±1.4	88.0±0.9	88.7±1.1
LiLT	40.1±0.5	48.4±5.9	–	85.7±0.5	84.4±0.3	83.8±1.2
BROS	40.3±1.8	45.0±4.9	35.1±3.1	87.8±0.4	87.5±1.1	87.5±1.3
BERT	39.4±0.3	46.6±3.0	40.5±0.1	77.1±0.9	77.4±0.6	77.2±0.4

erage accuracy AA $= \frac{1}{T}\sum_j R[T,j]$ and backward transfer BWT $= \frac{1}{T-1}\sum_{j<T}(R[T,j] - R[j,j])$.

4. Two Causes of Forgetting

4.1. Cause 1: drift of the output-facing computation

Forgetting is total within one boundary, on every backbone. Under naive sequential fine-tuning on the stream FUNSD→CORD→SROIE the first task collapses after one further task: FUNSD falls from 89.1 to 3.3 on LayoutLMv3, 83.9 to 2.1 on LiLT and 88.9 to 6.6 on BROS. Table 1 gives the DIL picture on four backbones.

Where it sits. Representations drift cumulatively with depth: per-token CKA [15] between consecutive checkpoints is 1.00 at the embeddings, 0.78 over the early layers, 0.34 in the middle, 0.25 at the last layer and 0.16 at the classifier, with the same profile on all four backbones (Fig. 2, left) and under LayoutLMv3 modality masks the full, text+layout and text-only models forget comparably (BWT −89.8 and −87.2 over three seeds, −79.1 over the two live text-only seeds). The classifier is also where parameters that mattered for the old task are overwritten: old-task-Fisher-weighted displacement is two to three orders of magnitude larger there than in any encoder bucket of the document encoders. That weighting favours the head by construction, and inside the encoder the weighted displacement sits in the early and middle layers, not the late ones, so we read "late layers" as a statement about representations only (supplement, Sec. D). This agrees with the image-classification anatomy of Ramasesh et al. [30] and is not our contribution; the interventions below are.

Freezing relocates the drift. Freezing the whole network after the first task gives AA 41.1 with BWT 0.0 on DIL: nothing is forgotten and nothing is learned. Freezing only the head and late layers, with the early and middle encoder left plastic, gives AA 38.8 and BWT −76.8, the naive level, while early-layer CKA, ≈1.0 under naive training, falls to 0.19. The drift moved into whatever remained plastic (Fig. 2, right; seed 42).

Only old-task supervision of the readout repairs it. On average replay comes within one point of joint training, or exceeds it, on all four backbones; EWC and LwF stay at the naive level (Tab. 1). We then built or ported five families that keep updating a shared network with no stored old-task targets (head merging, merging with Gaussian classifier realignment [35], subspace transfer, isolated head slots, feature-Gaussian replay): head merging is a no-op (21.6 against 21.9 unmerged at its operating point), merging with realignment reaches 43.9, subspace transfer 39.8, head slots 42.2, and feature-Gaussian replay 66.0 by recovering the densely labelled task only, against 41.3 for naive and 88 for replay (supplement, Sec. C). Where we measured the mechanism, on naive training and LwF, the head's output marginal on old tasks snaps to the just-trained task's marginal (cosine 0.997–0.998 against 0.60 to the old task's own), and classes that are never re-exercised fall to exactly 0.0 F1. Five post-hoc corrections of that marginal, each with a pass bar fixed before the run, recover at most −0.1 AA, because old-class probability mass is $\sim 10^{-6}$. Our claim is scoped accordingly: on a fixed label space, no method we tried that updates a shared trunk and readout without old-task supervision recovers the old tasks. Section 5.3 shows the claim does not extend to label growth, where isolation becomes possible.

The size of a remedy's benefit is backbone-dependent; the cause is not. The gap between joint and naive training on Mixed agrees to 2.5 points across the four backbones, whereas the benefit of EWC over naive spans 17 points and changes sign (−11.9 on LayoutLMv3, −2.6 on LiLT, +3.5 on BROS, +5.2 on BERT; legacy scoring, supplement Sec. D). A single-backbone comparison of methods without old-task supervision can land on either side of zero, which is why Tab. 1 reports four.

4.2. Cause 2: background conflict under label growth

On CIL-CORD under legacy scoring naive training, LwF, ER and DER++ sit at 16–19 AA against 33 for joint training, and several baselines collapse below that band; this has been read as a floor. Session-restricted scoring removes the artefact for the reference, which rises to 91.9±1.3, and leaves the methods far below it: ER 24.3±2.4, CoLaR 30.4, naive 18.9±0.3 (Tab. 3). The gap that remains is real, and it is not a memory problem.

A document stored at session t carries "O" on every entity introduced after t, and a current document carries "O" on every entity of earlier sessions. With one shared softmax both sets of labels supervise the same background logit, on receipts that look alike, in opposite directions. Improving the memory does not remove the contradiction: refreshing the stored labels with the true current labels, a ceiling no deployable method can reach, gives 61.2 (seed 42), and



Figure 2. **Drift is output-facing, and freezing moves it.** Left: linear CKA between checkpoints before and after the second task under naive fine-tuning falls with depth on all four backbones (1.0 = no drift; mean over three seeds). Right: freezing the head and late layers leaves backward transfer unchanged, (a) whatever the number of frozen layers and (b) on three DIL task orders and a ten-session CIL-CORD stream (LayoutLMv3, seed 42, no memory).

Table 2. **Memory × readout on CIL-CORD** (LayoutLMv3, AA, session-restricted scoring; mean ± s.d. over three seeds, per-seed values when fewer; †legacy protocol, settings per cell in the supplement). Rows change how drift is handled, columns how the background is supervised. The third row uses per-session low-rank adapters and no memory, but the two cells are different adapter methods, not one design with the readout swapped: CL-LoRA [9] (rank-8 orthogonal adapters on the attention projections, all applied together at inference, distillation loss, base trained, fp16 or fp32 by seed, 30-epoch cap) with a shared softmax, and SBR-Recall (rank-64 deltas on attention and feed-forward layers above k, one per block, base frozen after the first session, fp32, 10-epoch cap) with the blocked readout.

	Shared softmax	**Session-blocked readout**
No memory	18.9±0.3†	20.4†
Whole-document memory (CoLaR)	28.8/31.9†	**88.0±2.0**†
Per-session low-rank adapters, no memory	18.7/18.6†	**92.8±0.5**†
Joint training (reference)	91.9±1.3†	

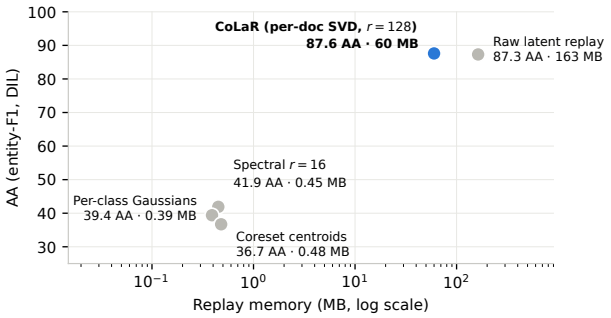


Figure 3. Memory size versus retention (DIL, LayoutLMv3, AA, seed 42; log-scale memory). Per-document SVD (CoLaR) matches raw latent replay at 2.7× less storage; every summary that breaks whole-document binding falls to the naive level.

restricting the replay loss to the classes alive at storage time gives 42.0 (two seeds). Perfect memory labels recover about half of the gap, because current documents still mislabel old entities. The continual-NER methods designed for the shift of "O" do not recover it either (Tab. 3).

4.3. The two causes separate

Table 2 crosses the remedy for each cause. Without a memory, blocking the readout barely helps (18.9 to 20.4): the labels are consistent but the trunk drifts. With a whole-document memory and a shared softmax, drift is contained but the labels conflict (30.4). Only the cell with both approaches joint training (88.0±2.0 against 91.9±1.3). The third row replaces the memory by per-session low-rank adapters: CL-LoRA, with a shared softmax, collapses (18.6, although its sessions reach 81 to 95 F1 when they are learned), and SBR-Recall, with a blocked readout, reaches 92.8±0.5. The two adapter methods also differ in rank, placement and training (Tab. 2), so this row contrasts adapters with a shared softmax against adapters with a blocked readout rather than isolating the readout alone. Drift can be handled by memory or by per-session adapters; in every cell we ran, the background conflict was resolved only at the readout.

5. Remedies

5.1. CoLaR: a memory for drift

A memory repairs drift by re-exercising the readout on inputs that keep their old labelling. To find the least it must store we vary the stored object and hold the mechanism fixed: activations are captured at the output of layer k, layers $1..k$ are frozen after the first task, and stored activations are replayed through the plastic upper layers and head (DIL, LayoutLMv3; single-seed diagnostics unless stated).

Whole documents retain; summaries collapse. Replaying the stored activations of 50 whole documents per task retains AA 87.3. Summaries of the *same* store collapse to the naive level: a rank-16 pooled spectral summary 41.9, per-class Gaussians 39.4, real k-means centroids 36.7–39.7. A single-variable control isolates the ingredient: four whole documents retain 63.8, and the identical token features placed on layout carriers from other documents retain 36.7.

The binding of a token's feature to its position and its label is worth 27 points. The requirement is specific to documents. On ViT-B/16 image class-incremental learning (CIFAR-100, ImageNet-R; supplement, Secs. B and C) a class-conditional Gaussian head (SLCA [47], 87.5/65.0) beats the same latent memory (80.0/55.1), and a ridge readout refit on the trained trunk reaches 88.3/72.3: there the trunk is intact and only readout statistics are stale. An image carries one label, so a per-class statistic summarises its readout; a document carries one label per token bound to a position, and the Gaussian analogue is null on documents in every form we tried (39.4 as a store).

Per-document compression. The pooled feature space is close to full rank, but one document's tokens share a layout and a vocabulary, so its activation matrix $H_j \in \mathbb{R}^{N_j \times d}$ (N_j tokens, width d) is low-rank. CoLaR (Compressed Latent Replay) stores a rank-r truncated SVD per document, $H_j \approx U_r \Sigma_r V_r^\top$, with the document's boxes, attention mask and labels, rebuilds $\hat{H}_j$ at replay, and injects it at layer k; the loss is cross-entropy on the current batch plus cross-entropy on replayed documents. At k=4, r=128 and 50 documents per task it retains 87.8±0.5 AA on DIL in 60 MB, against 87.9±1.4 for ER with 200 raw documents and 88.7±1.1 for joint training, and is 2.7× smaller than the uncompressed latent memory it matches (87.3 at 163 MB; Fig. 3); r=64 gives 80.6 at 32 MB. The memory holds no raw pixels or token ids. We make no formal privacy claim: mid-network activations can leak content. Rebalancing the memory (class-balanced loss, coverage-based selection, entity-weighted compression) trades one task for another without moving AA, and read-side additions are negative (supplement, Sec. C).

5.2. SBR: a readout for the background conflict

Let session b introduce the tag set $\mathcal{T}_b$. A shared softmax normalises over $\{\mathrm{O}\} \cup \bigcup_b \mathcal{T}_b$. The Session-Blocked Readout (SBR) gives every session its own background logit $o_b(x) = w_b^\top h(x)$ on the token feature $h(x)$ and normalises each block separately:

$$p_b(\cdot \mid x) = \mathrm{softmax}\big([\, o_b(x),\, z_{\mathcal{T}_b}(x)\,]\big), \tag{1}$$

where $z(x)$ are the tag logits of the usual linear head and $z_{\mathcal{T}_b}(x)$ its entries for the tags in $\mathcal{T}_b$. Within block b a token labelled "O" in session-b data is truly background, so the block's supervision is never stale. For a token x with label y in a document of session s,

$$\mathcal{L}(x, y, s) = \sum_b m_{sb}\, \mathrm{CE}\big(p_b(\cdot \mid x),\, y_b\big), \qquad y_b = \begin{cases} y & y \in \mathcal{T}_b \\ \mathrm{O}_b & \text{otherwise,} \end{cases} \tag{2}$$

with one rule for the mask: m_{sb}=1 when session s annotates every tag of $\mathcal{T}_b$ (the block is supervised fully); m_{sb}=1 with y_b=O_b when b belongs to a different annotation schema, so none of its entities can occur in the document; and m_{sb}=0 otherwise, because the document's "O" labels say nothing about block b. Stored documents follow the rule of the session that stored them. A session that adds no tags creates no block, so on a fixed label space SBR has a single block.

Inference needs no task identity. Each block reports how strongly it claims the token, $e_b = 1 - p_b(\mathrm{O}_b \mid x)$; the block with the largest e_b labels the token with its best tag if $e_b > 0.5$, and the token is "O" otherwise. The trunk is shared across blocks and kept stable by the CoLaR memory.

5.3. SBR-Recall: isolation in place of memory

With SBR the memory is written during training and never read at test time. SBR-Recall also stores, for each session $b > 1$, a low-rank delta Δ_b [11] of the layers above k (rank 64 on the attention projections and feed-forward layers), freezes these layers after the first session (the base θ_0; Δ_1=0), and computes block b under the trunk state that learned it:

$$p_b(\cdot \mid x) = \mathrm{softmax}\big([\, o_b,\, z_{\mathcal{T}_b}\,]\big)\big(x;\, \theta_0 + \Delta_b\big). \tag{3}$$

Every term of Eq. (2) is computed under the delta its block is read under, at training and at test time. Old blocks can then not drift, since neither the base nor their delta is updated, and routing by e_b is unchanged. Inference costs one pass through the upper layers per session.

SBR-Recall is the blocked-readout cell of the adapter row of Tab. 2. It works without a memory because label growth supplies what isolation normally lacks, a routing key: each session's background logit, trained on that session's true negatives, tells the model which delta a token belongs to. Where there is no such key, isolation does not replace memory. On a fixed label space there is one block and nothing to route by. Across schemas the all-background terms of Eq. (2) have no positive counterweight without stored documents: on Mixed, SBR-Recall without a memory scores 44.7 (seed 42), with other-schema blocks silent until their domain returns, and 85.9 with the 50-document memory restored.

6. Experiments

Main comparison. Table 3 reports the fixed comparison set. On CIL-CORD every method with a shared softmax stays at or below CoLaR (30.4), and the two blocked readouts close most or all of the gap: CoLaR-SBR 88.0±2.0 and SBR-Recall 92.8±0.5 against 91.9±1.3. On Mixed, CoLaR-SBR reaches 89.7 against 87.7 for joint training and 74.5 for ER. On DIL, where only drift is present and SBR has one block, CoLaR, CoLaR-SBR and ER all come close to joint training on average (87.8, 86.2, 87.9 against 88.7);

Table 3. **Comparison on LayoutLMv3** (AA under session-restricted scoring on CIL-CORD and Mixed; mean ± s.d. over three seeds, per-seed values when fewer; "–": not run). Full-label F1 is the final model's F1 on the unmasked CORD test set, recorded by clean-protocol runs only; $^{n=k}$: k clean seeds so far. ER keeps 200 raw documents in † cells and as many as CoLaR in clean cells. LexSlot: extra head slots selected by OCR-vocabulary overlap (see Ablations). †legacy protocol (test-split early stopping except CIL-CORD and Mixed joint training; epoch cap and precision differ by row, see Sec. 3). ^{c}LwF only: one fp16 seed collapsed to all-background (CIL-CORD seed 123, Mixed seed 7) and is excluded; LwF does not collapse in fp32. CPFD and IS3 keep their zero seeds, which are not precision effects (CPFD also collapses in fp32; IS3 always ran in fp32). ^{b}with the 50-document memory; 44.7 without.

	CIL-CORD		Mixed	DIL
Method	AA	full-label F1	AA	AA
Naive	18.9±0.3†	–	44.4/43.7†	41.3±1.5†
ER	24.3±2.4†	–	75.6/73.3†	87.9±1.4†
LwF	18.4/18.5†,c	–	47.7†,c	43.7±1.1†
DER++	20.4±1.4†	–	75.1/74.0†	88.0±0.9†
CODA-Prompt	14.0/15.9†	–	12.1/10.6†	32.5±1.2†
CL-LoRA (2025)	18.7/18.6†	–	45.7/43.9†	41.1±0.3†
CPFD (2023)	2.7/0.0†	–	0.3/15.1†	6.1±7.3†
IS3 (2024)	5.6/7.7†	–	0.0/40.6†	38.3†
LCA (2026)	13.7/7.0†	–	38.0/36.4†	43.9±2.5†
CoLaR	28.8/31.9†	–	75.6/74.6†	87.8±0.5†
CoLaR-SBR	88.0±2.0†	88.8$^{n=1}$	**89.8**/**89.6**†	86.2±1.3†
+ LexSlot head slots	89.7±0.5†	–	89.6±1.0†	87.0±0.4†
SBR-Recall	**92.8±0.5**†	–	85.9†,b	–
Joint training (reference)	91.9±1.3†	–	86.9/88.5†	88.7±1.1†

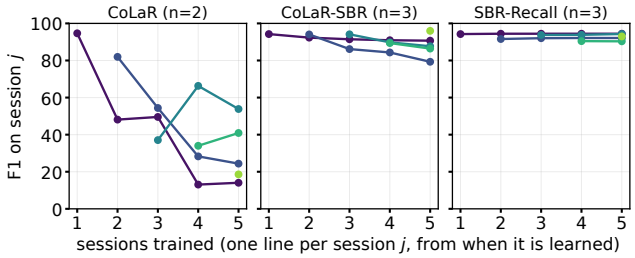


Figure 4. **Retention per session on CIL-CORD** (LayoutLMv3, session-restricted scoring; F1 on session j after each later session, mean over the n seeds given in each panel title). With a shared softmax every session decays as soon as the next arrives; the blocked readout holds them, with a slow loss on the earliest sessions; under SBR-Recall every curve is flat.

SBR (86.2) is below CoLaR (87.8) there, and its only differences are the separate background logit and the thresholded decision.

Ablations. The blocked readout on a trunk frozen after the first session scores 14.8 (later sessions cannot be learned), and without a memory 20.4, so neither the readout nor plasticity alone is the effect. Tuning the claim threshold and a per-block bias on the test set itself changes AA by at most 0.6 (seed 42), so what SBR still loses to joint training is retention of early sessions (Fig. 4), which is the part SBR-Recall removes. For SBR-Recall without a memory, rank 8 on the attention projections alone gives 82.5 (seed 42) with flat retention but two sessions stuck at 65 and 68 F1: isolation is sound and the capacity of the delta is what CORD's look-alike price classes need. Stacking LexSlot on CoLaR-SBR (extra head slots that a session updates only when its documents' OCR vocabulary selects them) leaves the mean within two points on all three scenarios and narrows the spread between seeds on CIL-CORD and DIL; both mechanisms partition the same readout. Unlike CoLaR, LexSlot stores the token ids of its documents.

Cost. CoLaR-SBR stores 50 compressed documents per session (96 MB on CIL-CORD) and one background logit per session. SBR-Recall stores no documents on a single-schema stream and 54 MB of half-precision deltas for the four later sessions, and pays one upper-layer pass per session at inference (about 3.7× the compute of one forward pass at five sessions and k=4).

7. Limitations

The study covers three English datasets with test sets of 50 to 347 documents and four encoder-only backbones; SBR and SBR-Recall are evaluated on LayoutLMv3. Cells marked † early-stop on the test split (except CIL-CORD and Mixed joint training) and are not compared against clean-protocol cells in the text. The freezing experiment and the memory-content ladder are single-seed diagnostics; the tables are three-seed unless a cell shows fewer values. Joint training is a reference, not a bound: with early stopping it can be matched or slightly exceeded. SBR-Recall grows by one delta and one inference pass per session, freezes its base on the first session, so its accuracy can depend on which session comes first, and does not replace memory on a fixed label space or across annotation schemas; its Mixed result is one seed. SBR's supervision rule uses the annotation schema of each session, which is known at training time and is not used at test time. CoLaR is a replay method whose memory grows with the number of sessions; its settings were chosen on LayoutLMv3 and transfer to BROS but trail raw replay by 5–9 points on average on the text-heavy BERT and LiLT. CoLaR stores no raw pixels or token ids, which is weaker than privacy.

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